

ENGINEERING PROJECT PORTFOLIO



CHIMWEMWE CHINKUYU

B.S.E. Mechanical & Aerospace Engineering · Minor in Computer Science · Princeton University, Class of 2026

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EDUCATION

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MOTIVATION

I'm inspired by the possibility to turn ideas into smart, functional solutions at the intersection of hardware and software. My work spans mechanical design, fabrication, and software development.

I'm eager to grow in innovation-focused environments and contribute to advanced devices, autonomous systems, and technologies that enhance human health and quality of life.



TECHNICAL SKILLS

CAD / CAM

- PTC Creo (incl. FEA)
- Fusion 360
- Onshape
- AutoCAD

Programming & Robotics

- Python
- MATLAB
- Java
- C
- ROS
- COMPAS FAB
- UR-RTDE
- PID control

Fabrication

- 3D Printing (FDM, SLA, SLS)
- Manual & CNC Machining
- Carpentry
- Robotic Fabrication
- Autonomous Assembly

Software & Tools

- Simulink
- Jupyter Notebook
- Docker
- Anaconda
- Grasshopper
- NVIDIA Jetson
- RadiAnt DICOM Viewer

LINKEDIN

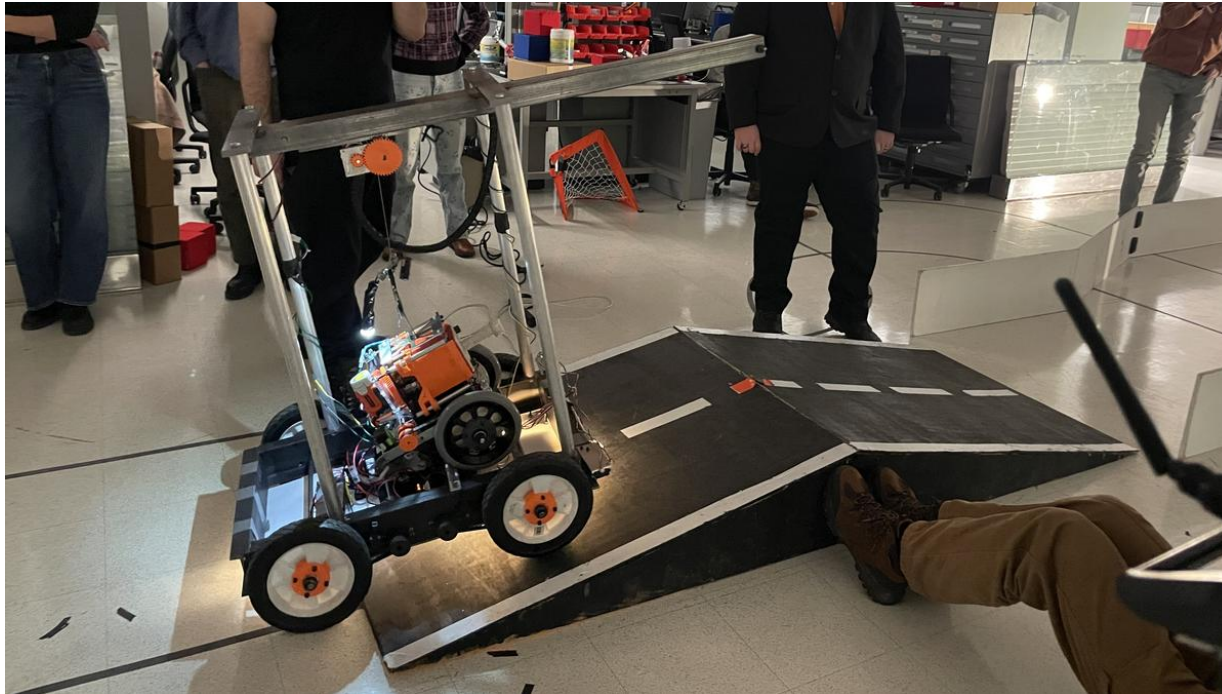
[linkedin.com/in/chimwemwe-chinkuyu](https://www.linkedin.com/in/chimwemwe-chinkuyu)

PORTFOLIO WEBSITE

ckcgithub16.github.io/CC_Portfolio/

Design & Manufacturing of a Search and Rescue Robot

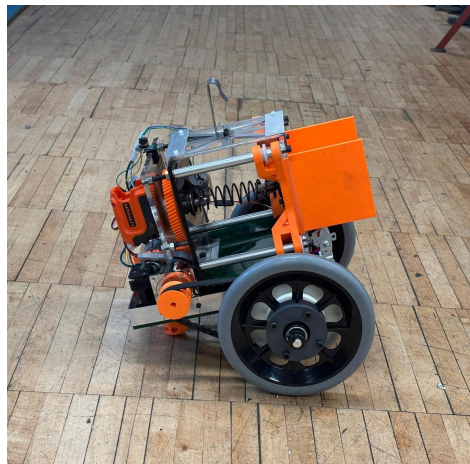
Oct – Dec 2025



SARR traveling over the ramp



Motherbot deploying Babybot over the wall



The Babybot

OVERVIEW

For a mechanical design course, I collaborated with 11 classmates to build a marsupial search and rescue robot that could navigate an obstacle course to deliver a med kit. The system pairs a large Motherbot that carries a smaller Babybot onboard. The Motherbot traversed a ramp, a chute, and a 36-inch wall. At the wall, a motorized lifting arm raised the Babybot over the top, deploying it on the far side. The Babybot then navigated independently to a target zone and delivered a med kit payload.

SENSING & CONTROL

- Both robots used C++ state machine controllers on Teensy microcontrollers.
- The Motherbot used CdS phototransistors for line-following — proportional on flat ground, bang-bang on the ramp, PD in the chute — and ultrasonic sensors to trigger state transitions.
- The arm deployment was open-loop.
- The Babybot used similar sensing for post-deployment navigation.

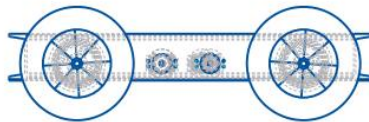
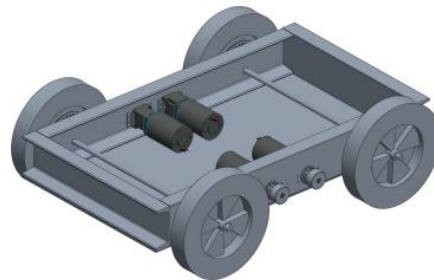
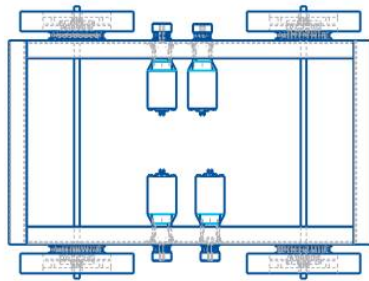
RESULT

- The Motherbot navigated the entire course, deployed the Babybot over the 36-inch wall, and the Babybot delivered the med kit to the target zone at the class final.

Search and Rescue Robot — Chassis Design & Manufacturing

Oct – Dec 2025

Motherbot Chassis Assembly



MY CONTRIBUTION

- Designed the low-carbon steel chassis, 60:1 two-stage HTD belt drivetrain, and engineering drawings in Creo Parametric.
- Led fabrication of the chassis and drivetrain.

CHASSIS DESIGN

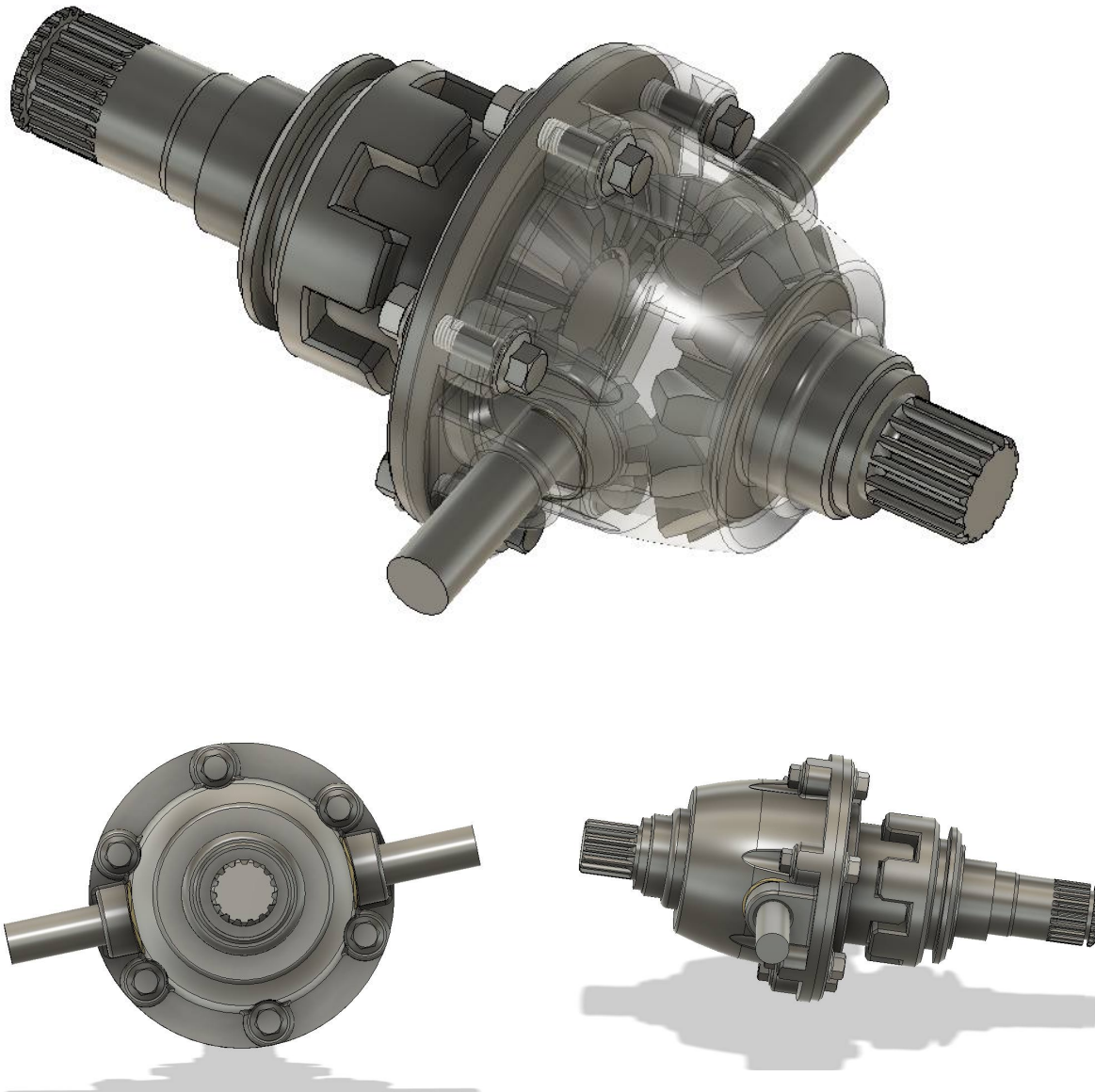
- AISI 1018 steel was chosen for weldability and stiffness.
- FBD analysis confirmed tip-over stability under full arm extension.
- FEA at motor mount positions and the lifting arm pivot verified structural margins.
- The two-stage belt drivetrain over a jackshaft provided torque for the ramp; slotted motor mount holes allowed precise belt tensioning.

MANUFACTURING

- Steel frame members were band-saw cut, cope-cut on a knee mill, and MIG welded.
- Motor mount brackets and the jackshaft were manually machined on the lathe and knee mill.
- Sensor mounts and the lifting arm cradle were FDM 3D printed for rapid iteration.
- Key drivetrain components were CNC machined for dimensional accuracy.

Custom Differential Gearbox

June 2025



OVERVIEW

As a personal project to sharpen my CAD skills, I designed a fully parametric bevel-gear axle differential entirely in Autodesk Fusion 360. The planetary differential configuration distributes drive torque between two output shafts while allowing independent rotation speeds, enabling smooth, skid-free turning.

DESIGN APPROACH

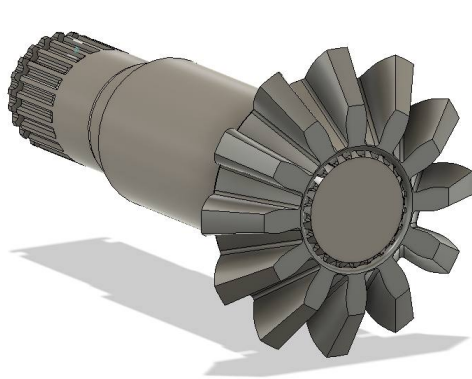
- Gear geometry was determined using standard bevel gear formulas driven by linked Fusion 360 equations.
- Module, pressure angle, tooth count, and pitch circle diameter update the full geometry automatically as inputs change.

SKILLS USED

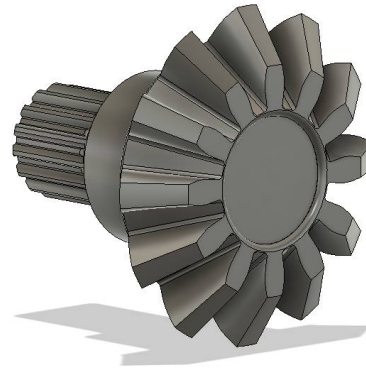
Autodesk Fusion 360 · Parametric Modeling · Bevel Gear Theory

Custom Differential Gearbox — Component Drawings

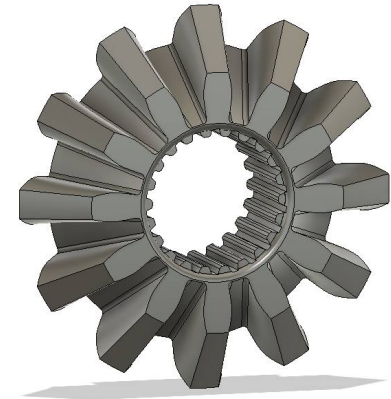
June 2025



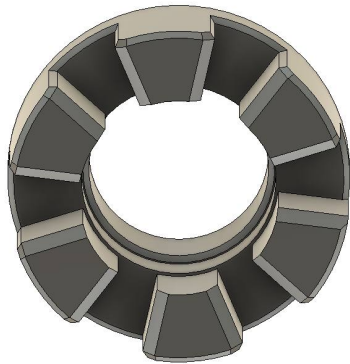
Long Drive Shaft



Short Drive Shaft



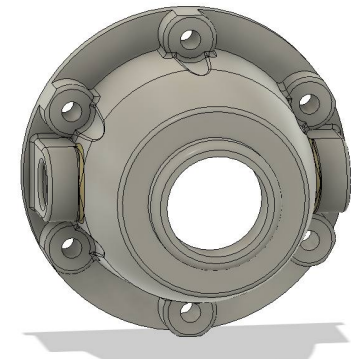
Bevel Gear



Differential Ring



Output Hub



Outer Casing

Self-Driving RC Truck with Obstacle Avoidance & Safety Filter

Mar – May 2025



Navigating the Task 1 obstacle course

■ [Watch the team's Task 1 run on YouTube](#)

OVERVIEW

For an intelligent robotics course, I worked in a team of 4 to program an autonomous RC truck, powered by an NVIDIA Jetson, to complete two tasks: racing through an obstacle course, and running a real-time driver-assist safety filter.

MY CONTRIBUTION

Algorithm development · Code implementation · Debugging of both software and hardware

TASK 1 — OBSTACLE COURSE RACING

- Implemented an iLQR (iterative Linear Quadratic Regulator) controller with cost-prioritization to navigate 16 ordered waypoints in 3:30 while avoiding up to 20 obstacle cubes.
- Finished 4th fastest of 14 teams after time penalties.

TASK 2 — SAFETY FILTER

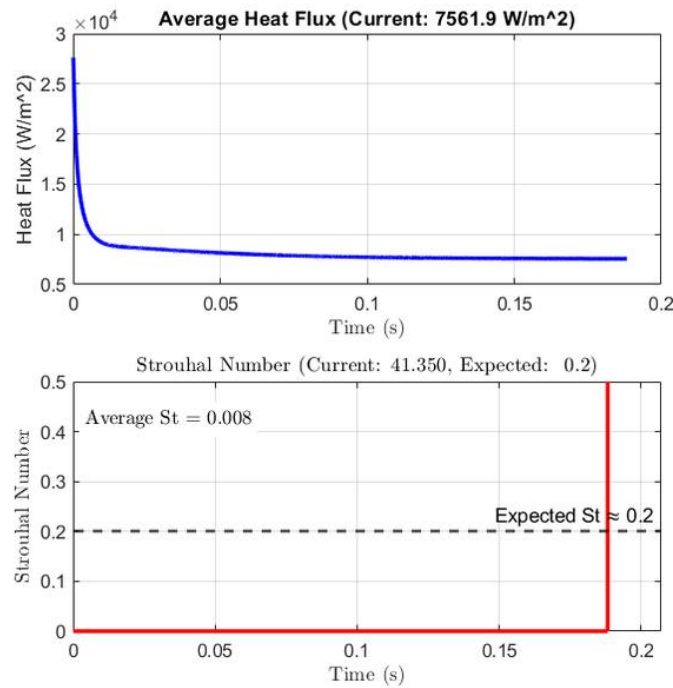
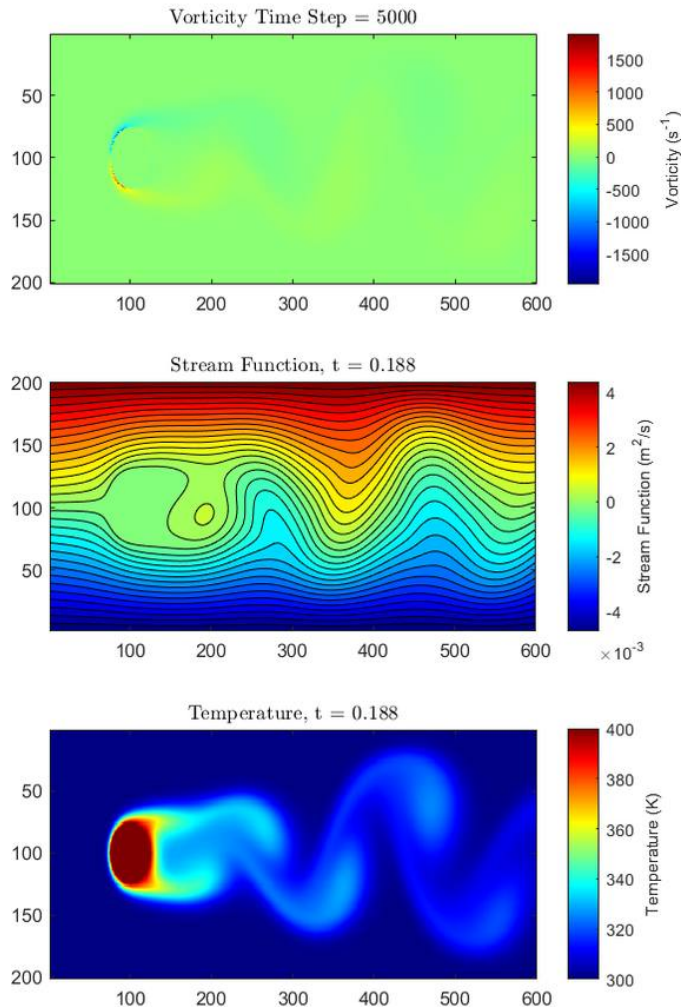
- Built an Advanced Driver Assistance System (ADAS) that uses iLQR to monitor manual driver inputs in real time.
- Overrides steering that would leave the track or cross lane boundaries while minimizing unnecessary interventions.

SKILLS USED

Python · ROS · Optimal Control · Path Planning · Obstacle Avoidance

2D Unsteady Heat Transfer: Heated Cylinder & NACA 2412 Airfoil

Apr – May 2025



OVERVIEW

For a heat transfer course, I built a Psi-Omega (streamfunction-vorticity) finite-difference simulation from scratch in MATLAB to model 2D unsteady heat transfer over two geometries: a 400°C cylinder in 300°C flow (convective heat flux 7.6 kW/m²), and a NACA 2412 airfoil at 5° and 30° angles of attack (heat flux 1,002.6 vs 1,133.6 W/m²).

TECHNICAL APPROACH

- Three coupled PDEs solved at each time step:
- Streamfunction (Poisson) via Gauss-Seidel with over-relaxation
- Vorticity (momentum) via forward Euler with upwind differencing
- Temperature (energy) coupled to the velocity field

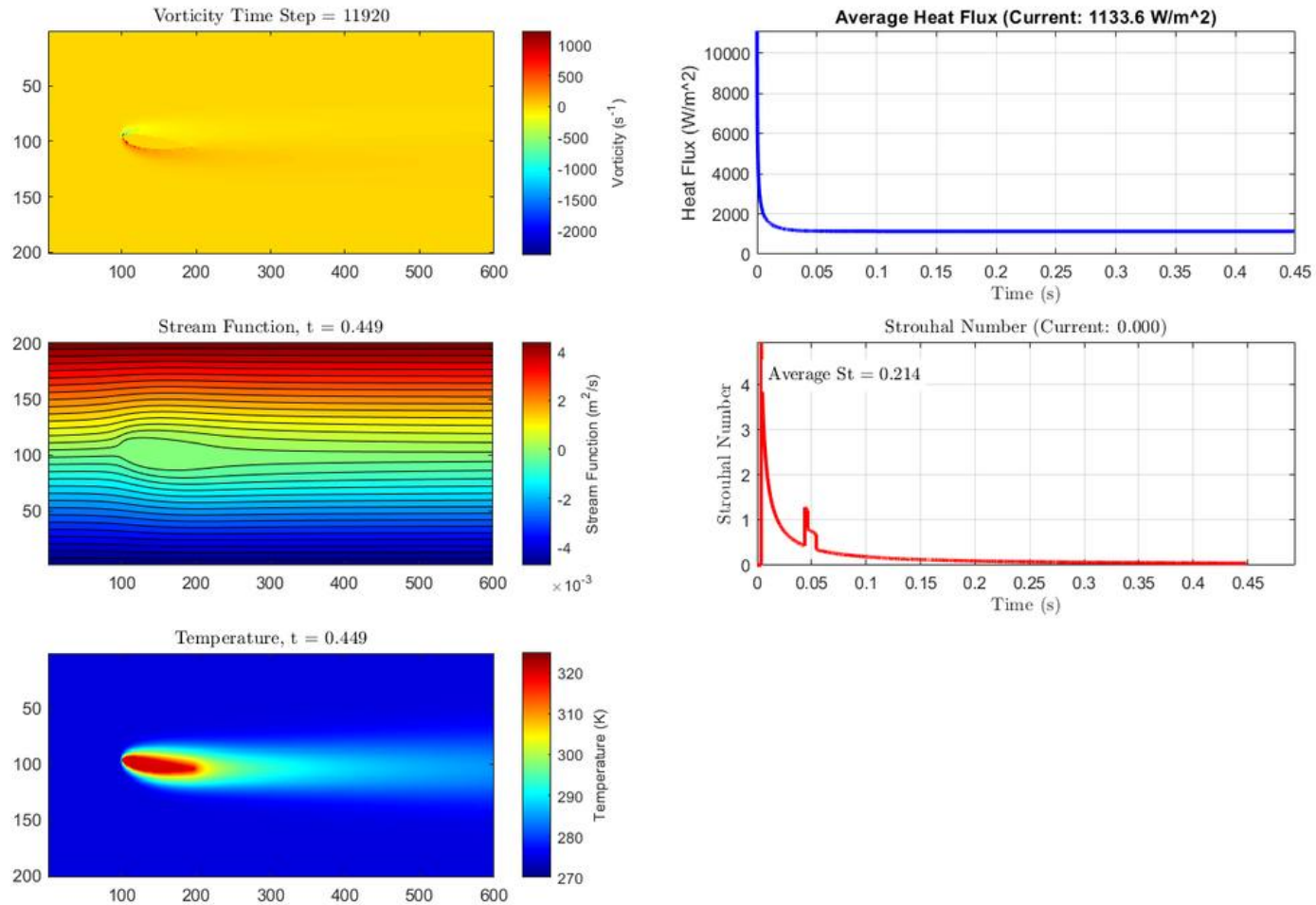
SKILLS USED

MATLAB · Finite Difference Methods
 · Fluid Dynamics · Heat Transfer ·
 Numerical Stability · CFD Validation

Heated cylinder — vorticity, stream function, temperature, and convergence history

Heat Transfer Simulation — NACA 2412 at 5° Angle of Attack

Apr – May 2025

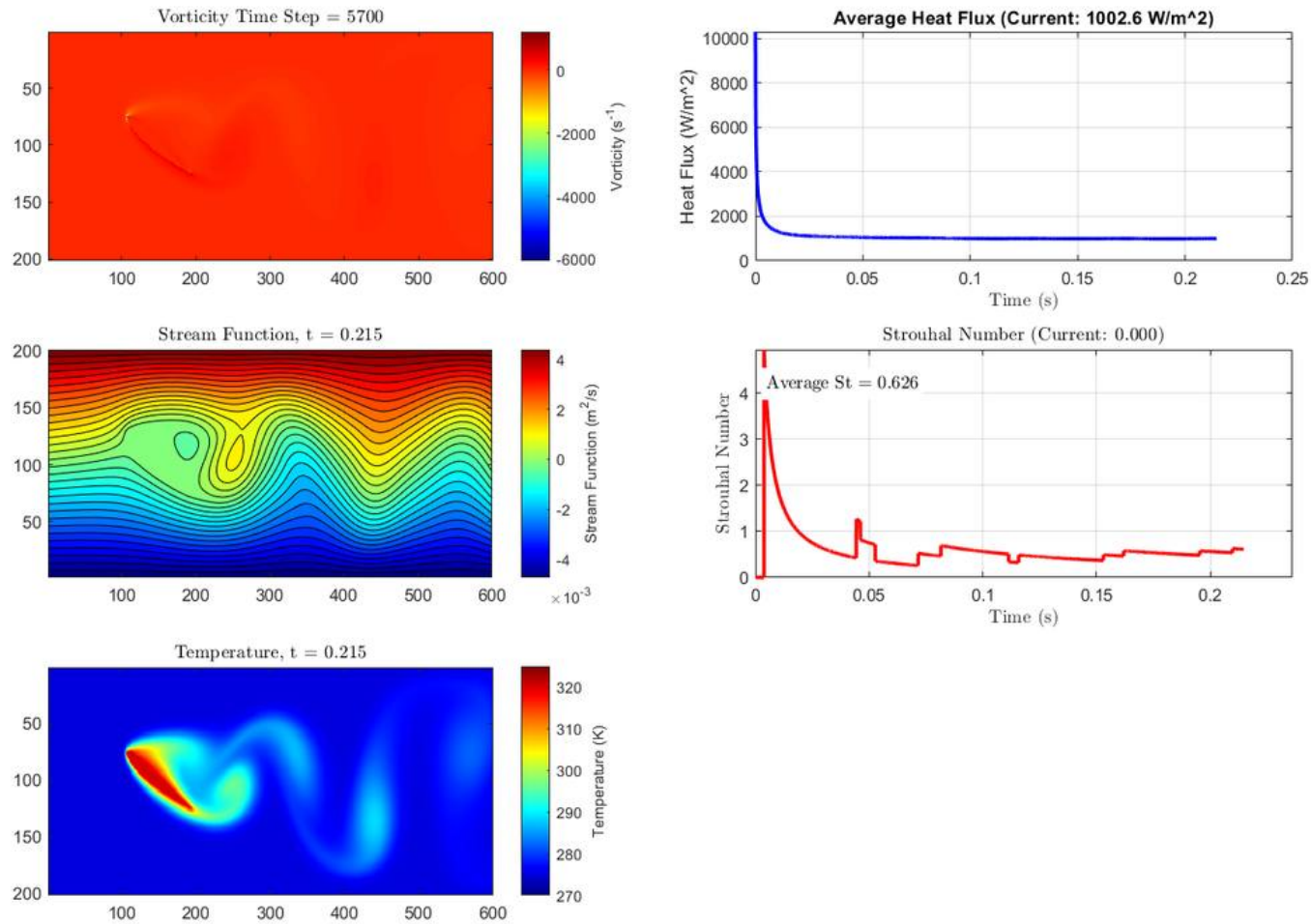


ATTACHED FLOW

Flow remains attached and the temperature distribution is smooth and symmetric about the chord line. The result was validated against SimFlow CFD — within 5% of the commercial result — confirming correct PDE discretization and boundary conditions.

Heat Transfer Simulation — NACA 2412 at 30° Angle of Attack

Apr – May 2025

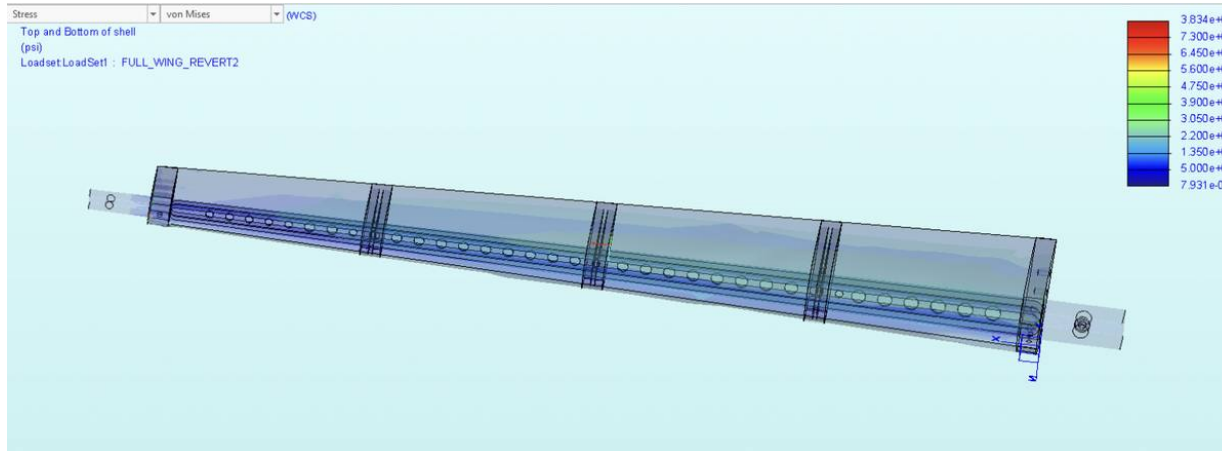


SEPARATED FLOW & VORTEX SHEDDING

Flow separates at the leading edge, producing large unsteady vortices shed periodically into the wake, with Strouhal number $St = 0.62$ — consistent with separated-flow theory. Higher stagnation temperatures and increased heat flux (13% above the 5° case) result from the thickened boundary layer.

Design and Manufacturing of a Load-Bearing Airfoil

Mar – May 2023



Full-wing FEA — von Mises stress at 97 lbf

OVERVIEW

The goal was to design a wing that could carry a specified load while minimizing deflection, cost, and weight. The 9-member team's 25.5-inch airfoil supported 97 lbf at 23.5 in. from the secured end with 3.75 in. tip deflection, weighed 18.3 oz, and failed at 120 lbs (beam web torsion) — above the 97 lbf design load.

DESIGN

- Aluminum 7075-T651 I-beam (yield stress 73,000 psi) with circular lightening holes — selected for best strength-to-weight ratio from four competing beam concepts.
- Nylon bulkheads with circular cutouts reduced weight.
- Balsa wood skin formed the NACA profile.

FEA & VALIDATION

- PTC Creo FEA validated deflection at 97 lbf.
- Failure mode — beam web torsion collapse at 120 lbs — was not captured in the planar FEA model, teaching the importance of 3D torsional load modeling.

SKILLS USED

PTC Creo · FEA · CNC Milling (Haas VF-7/40) · Waterjet (ProtoMAX) · FDM & SLS 3D Printing · Structural Analysis · GD&T



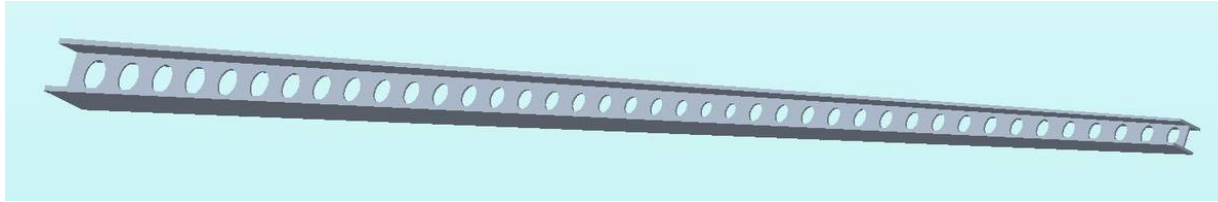
Wing under 97 lbf test load



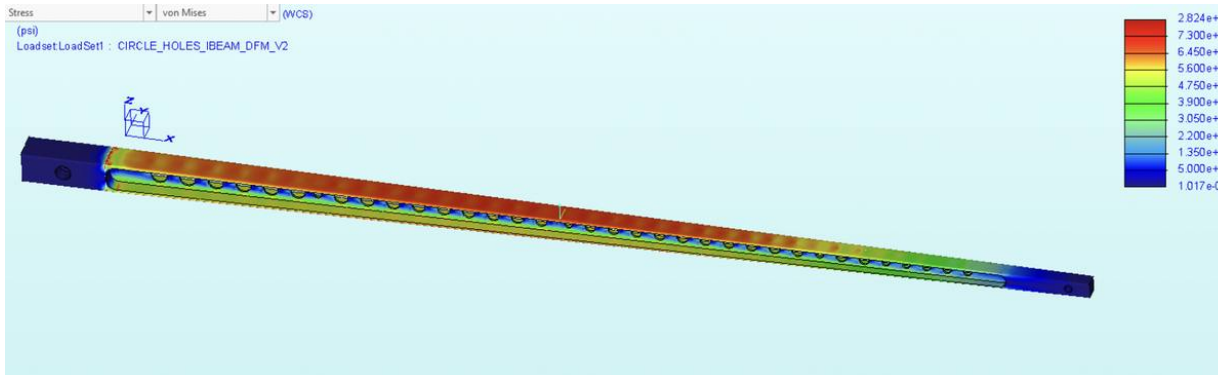
Completed wing assembly

Load-Bearing Airfoil — Manufacturing & Testing

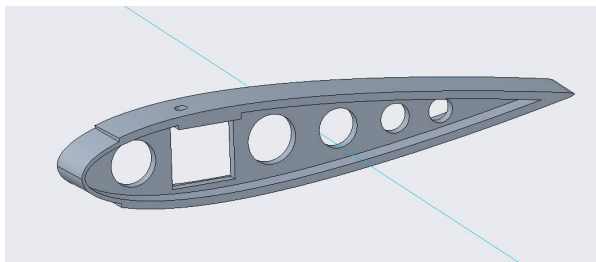
Mar – May 2023



Circular-Hole I-Beam



I-Beam FEA



Nylon Bulkhead Design



Waterjetted Bulkhead

MANUFACTURING

- I-beam CNC milled on Haas VF-7/40.
- Bulkheads waterjet-cut on ProtoMAX.
- Ribs printed on Ender 3 Pro (FDM) and structural parts on Formlabs SLS.
- Balsa skin cut on bandsaw and sanded to NACA profile.

TESTING RESULTS

- Passed the 97 lbf design load at 3.75 in. deflection, then continued loading to failure at 120 lbs.
- All weight targets were met.

LESSONS LEARNED

- Torsional loads must be explicitly modeled in 3D FEA.
- Physical load paths in complex assemblies differ from simplified beam theory predictions, especially near joints and web openings.